

Anthony M. Alam

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EDUCATION

University of Ottawa

Bachelor of Applied Science in Software Engineering - GPA: 3.96/4.00

Ottawa, Ontario

Graduation: December 2027

EXPERIENCE

Cisco Systems

Ottawa, Ontario

Incoming Software Engineer Intern (Telemetry and Observability)

Summer 2026

- Designing agentic AI workflows to automate observability tasks across distributed routing systems, built on model-driven programmability foundations (YANG, gNXI, NETCONF, RESTCONF).

Software Engineer Intern (IOS XE)

January 2025 - April 2025

- Led the development of a Python (Flask) code injection tool adopted by 75+ engineers, allowing previously unreachable conditions to be tested and increasing file line and branch coverage by 22% and 13% on average.
- Architected and re-implemented an optimal ConditionBlock graph data structure to support DAG-based flow analysis of C code, improving run time by 18% and cutting memory usage by 20%.
- Developed a library of 15 reusable APIs and algorithms for ConditionBlock graphs, already leveraged across 20+ Python files to support graph traversals, conditional path generation, and SQLite integration.
- Designed and implemented automated pre-commit routines with Python (pyATS) and YAML, covering all known edge cases and eliminating a 2-hour manual upgrade/downgrade process for Cisco Catalyst high availability switches.

Software Engineer Intern (IOS XR)

May 2024 - August 2024

- Led the development of an automated Python (pyATS) test suite for gRPC/gNMI requests, ensuring contract compliance in a multimillion-dollar customer deployment.
- Increased code coverage (+5% line, +4% branch) by developing unit tests in C (Cmocka), helping the team achieve its 80% coverage goal.
- Upgraded defect detection processes and the long-term maintainability of 100+ C files by integrating ASan and Valgrind analyses into nightly sanity runs.

Ford Motor Company

Ottawa, Ontario

Software Developer Intern

September 2025 - December 2025

- Spearheaded the development of a C++ SOA proxy service feature enabling hotspot credential read/write operations across 400,000+ next-generation Ford and Lincoln vehicles.
- Developed a custom parser built with regular expressions to identify Wi-Fi service failures across millions of lines of ADB device logs, enhancing efficiency by 24% compared to the previous implementation.

PROJECTS

Grant-Winning Accessibility Tool (for Rutgers University) | JavaScript, HTML, CSS, OpenAI API

[GitHub](#)

- Spearheaded the development of a full-stack chatbot that evaluates the accessibility of Rutgers University online courses, earning a grant for AI innovation from the University of Ottawa.
- Built a custom CSV parsing algorithm to process thousands of lines of course data, enabling faculty to upload course content and instantly receive tailored accessibility recommendations based on the WCAG 2.1 guidelines.

PlayItAgain: Catan Game Save iOS App | Swift, SwiftUI, Core Image, YOLOv8, Core ML

In Development

- Building an iOS app that digitizes physical Catan boards, cards, and game state from a photo, allowing players to save and resume games across sessions.
- Designing a Core Image perspective-correction pipeline and an on-device YOLOv8 + Core ML detector to recognize terrains, number tokens, and player pieces.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, SQL, Bash, YAML, HTML, CSS

Frameworks & Tools: Flask, React, PostgreSQL, NoSQL, pyATS, JUnit, Cmocka, Git, GitHub, Docker, CI/CD, REST API, Unix/Linux (RHEL), VS Code, IntelliJ, Eclipse, Android Studio, Copilot, Claude Code, Jenkins, AWS

Engineering Skills: Full-Stack Development, Automation Testing, OOP, Networking (HTTP, IP, TCP, UDP), SDLC, Distributed Systems, Cloud Computing, Debugging (GDB), Code Review, Agile Development, Complexity Analysis

Relevant Coursework: Data Structures & Algorithms, Software Engineering Design Patterns, Operating Systems